

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

Overview of Internet

- 1 Networks
- 2 Switching
- 3 The Internet
- 4 Accessing the Internet
- 5 Hardware and Software



Overview of Internet

- 1 **Networks**
- 2 Switching
- 3 The Internet
- 4 Accessing the Internet
- 5 Hardware and Software



Networks

- A network is the interconnection of a set of devices capable of communication.
- A device can be a **host** (or **end system**) such as:
 - Large computer
 - Desktop
 - Laptop
 - Workstation
 - Cellular phone
 - Security system
- A device can also be a connecting device such as:
 - Router – connects the network to other networks
 - Switch – connects devices together
 - Modem (Modulator-Demodulator) – changes the form of data
- Devices in a network are connected using:
 - Wired transmission media (cables)
 - Wireless transmission media (air)
- Connecting two computers at home using a plug-and-play router creates a small network.

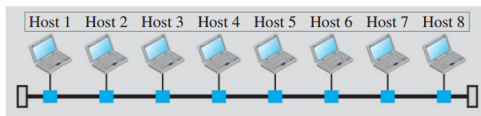
Local Area Network (LAN)

- A **Local Area Network (LAN)** is usually privately owned and connects hosts within:
 - A single office
 - A building
 - A campus
- Depending on organizational needs, a LAN can be:
 - As simple as two PCs and a printer in a home office
 - Large enough to extend throughout a company
 - Include audio and video devices
- Each host in a LAN has a unique identifier called an **address**.
- A packet sent from one host to another contains:
 - Source host address
 - Destination host address
- In earlier LANs:
 - All hosts were connected using a common cable
 - A packet sent by one host was received by all hosts
 - Intended recipient accepted the packet
 - Other hosts discarded the packet

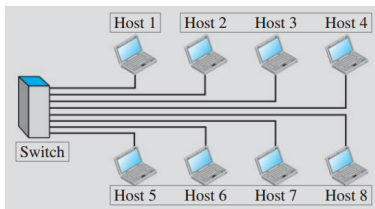
Local Area Network (LAN)

- Modern LANs use a **smart connecting switch**:
 - Recognizes the destination address
 - Sends packets only to the intended destination
 - Avoids sending packets to all hosts
- Advantages of switches:
 - Reduces traffic in the LAN
 - Allows multiple pairs of hosts to communicate simultaneously
 - Simultaneous communication is possible when there is no common source and destination
- A LAN does not define any minimum or maximum number of hosts.
- Figure 1.1 shows a LAN using:
 - A common cable
 - A switch

LAN Diagram



a. LAN with a common cable (past)



b. LAN with a switch (today)

Legend

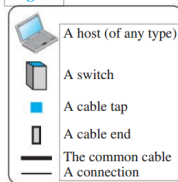


Figure: Local Area Network using Switch

Wide Area Network (WAN) - 1

- A **Wide Area Network (WAN)** is an interconnection of devices capable of communication.
- A **LAN** is limited in size:
 - Office
 - Building
 - Campus
- A **WAN** covers a wider geographical area:
 - Town
 - State
 - Country
 - Entire world
- A LAN interconnects:
 - Hosts
- A WAN interconnects connecting devices such as:
 - Switches
 - Routers
 - Modems

Wide Area Network (WAN) - 2

- A LAN is normally:
 - Privately owned by the organization
- A WAN is normally:
 - Created and operated by communication companies
 - Leased by organizations
- Two distinct examples of WANs:
 - Point-to-Point WAN
 - Switched WAN

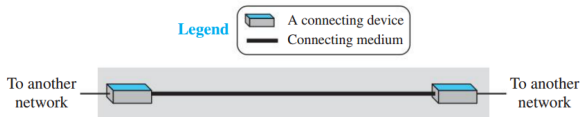
Difference Between LAN and WAN

Feature	LAN	WAN
Full Form	Local Area Network	Wide Area Network
Coverage Area	Office, Building, Campus	Town, State, Country, World
Connects	Hosts / End devices	Routers, Switches, Modems
Ownership	Privately owned	Communication companies
Speed	High speed	Comparatively slower
Cost	Low cost	High cost
Examples	Office network	Internet

Point-to-Point WAN

- A **Point-to-Point WAN** is a network that connects:
 - Two communicating devices
- The connection is established through transmission media such as:
 - Cable
 - Air (wireless medium)
- Point-to-Point WANs are used to connect networks to one another.
- Figure 1.2 shows an example of a Point-to-Point WAN.

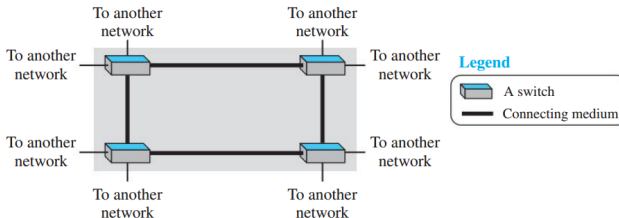
Figure 1.2 *A point-to-point WAN*



Switched WAN

- A **Switched WAN** is a network with:
 - More than two ends
- A Switched WAN is used in:
 - The backbone of global communication
- A Switched WAN is a combination of:
 - Several Point-to-Point WANs
 - Connected using switches
- Figure 1.3 shows an example of a Switched WAN.

Figure 1.3 A switched WAN



Internetwork - 1

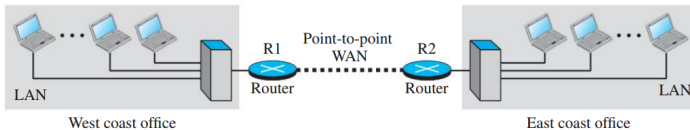
- Today, LANs and WANs are rarely used in isolation.
- When two or more networks are connected, they form:
 - Internetwork
 - Internet
- Example:
 - An organization has two offices:
 - East coast office
 - West coast office
- Each office has a LAN that allows communication among employees.
- To enable communication between offices:
 - A dedicated Point-to-Point WAN is leased
 - Service provider example: Telephone company

- The WAN connects the two LANs.
- The company now has:
 - An internetwork
 - A private internet (lowercase i)
- Communication between offices becomes possible.
- Figure 1.4 shows this internet.
- When a host sends a message within the same office:
 - Router blocks the message from leaving the LAN
 - Switch forwards the message to the destination

Internetwork - 3

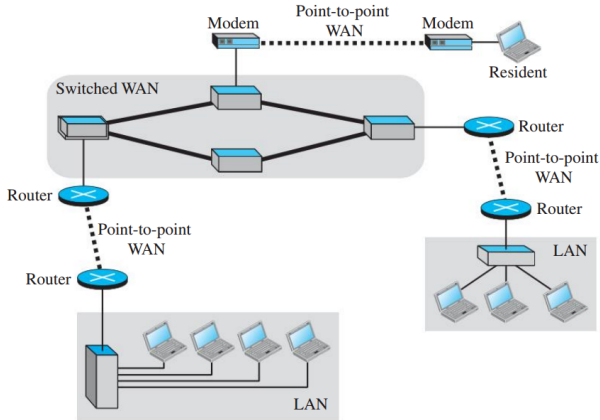
- When a host on the west coast sends a message to the east coast:
 - Router R1 routes the packet
 - Packet reaches Router R2
 - Packet reaches the destination host
- Figure 1.5 shows another internet with:
 - Several LANs
 - Several WANs
- One of the WANs is a switched WAN with:
 - Four switches

Figure 1.4 *An internetwork made of two LANs and one point-to-point WAN*



Internetwork - 3

Figure 1.5 *A heterogeneous network made of four WANs and three LANs*



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Switching

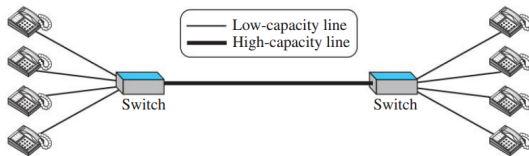
- An internet is a combination of:
 - Links
 - Switches
- Examples of switches:
 - Link-layer switches
 - Routers
- An internet is a **switched network** in which:
 - A switch connects at least two links together
- A switch forwards data:
 - From one link to another link
 - Whenever required
- Two common types of switched networks are:
 - Circuit-switched networks
 - Packet-switched networks

Circuit-Switched Network - 1

- In a **Circuit-Switched Network**, a dedicated connection called a **circuit** is always available between two end systems.
- The switch can only:
 - Make the circuit active
 - Make the circuit inactive
- Figure 1.6 shows a simple switched network connecting:
 - Four telephones on each side
- Telephone sets are used instead of computers because:
 - Circuit switching was common in telephone networks
- Today, part of the telephone network uses:
 - Packet-switched networks

Circuit-Switched Network

Figure 1.6 *A circuit-switched network*



Circuit-Switched Network - 2

- The four telephones on each side are connected to:
 - A switch
- The switch connects a telephone set at one side to a telephone set at the other side
- The thick line between switches is:
 - A high-capacity communication line
- The communication line can handle:
 - Four voice communications simultaneously
- The line capacity is shared among:
 - All telephone pairs

Circuit-Switched Network - 3

- The switches in this network:
 - Perform forwarding tasks
 - Do not have storing capability
- Case 1:
 - All telephone sets are busy
 - Four people communicate simultaneously
 - Full capacity of the communication line is used
- Case 2:
 - Only one telephone pair communicates
 - Only one-fourth of the capacity is used

Circuit-Switched Network - 4

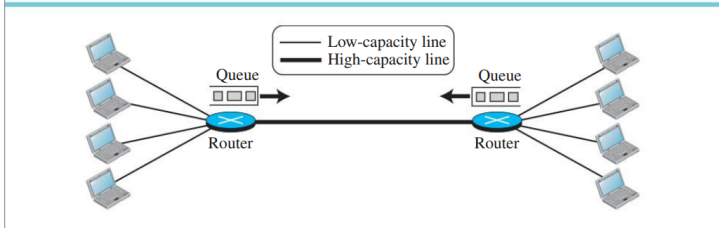
- Circuit-switched networks are efficient only when:
 - Working at full capacity
- Most of the time, the network is inefficient because:
 - It works at partial capacity
- The thick communication line has four times the capacity of a single voice line because:
 - Communication should not fail
 - All telephone sets may communicate simultaneously

Packet-Switched Network - 1

- In a computer network, communication is performed using:
 - Blocks of data called **packets**
- Unlike telephone communication:
 - Communication is not continuous
 - Individual packets are exchanged between computers
- Packet switching allows switches to:
 - Store packets
 - Forward packets
- A packet is:
 - An independent entity
 - Can be stored and sent later
- Figure 1.7 shows:
 - A packet-switched network
 - Four computers connected at each site

Packet-Switched Network

Figure 1.7 *A packet-switched network*



Packet-Switched Network - 2

- A router in a packet-switched network contains:
 - A queue
- The queue is used to:
 - Store packets
 - Forward packets
- Assume:
 - The thick line has twice the capacity of the data line
- If only two computers communicate:
 - Packets are transmitted without waiting



Packet-Switched Network - 3

- When packets arrive while the thick line is fully busy:
 - Packets are stored in the router queue
- Packets are forwarded:
 - In the order they arrived
- Packet-switched networks are:
 - More efficient than circuit-switched networks
- However, packets may experience:
 - Delays

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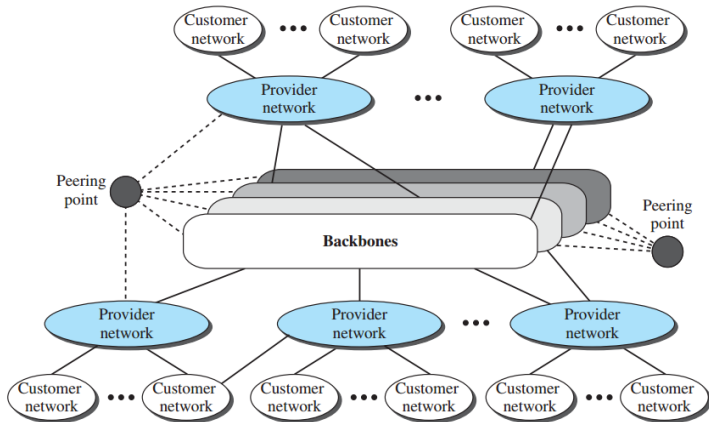


The Internet - 1

- An **internet** (lowercase i) is:
 - Two or more interconnected networks
 - Networks that can communicate with each other
- The most notable internet is the **Internet** (uppercase I).
- The Internet is composed of:
 - Thousands of interconnected networks
- Figure 1.8 shows:
 - A conceptual view of the Internet
 - Not a geographical view

The Internet

Figure 1.8 *The Internet today*



The Internet - 2

- The Internet consists of:
 - Backbone networks
 - Provider networks
 - Customer networks
- Backbone networks are:
 - Large networks
 - Owned by communication companies
- Examples of backbone companies:
 - Sprint
 - Verizon (MCI)
 - AT&T
 - NTT
- Backbone networks are connected through:
 - Peering points

The Internet - 3

- Provider networks are:
 - Smaller networks
 - Connected to backbone networks
- Provider networks:
 - Use backbone services for a fee
 - May connect to other provider networks
- Customer networks are:
 - Networks at the edge of the Internet
 - Actual users of Internet services
- Customer networks pay:
 - Fees to provider networks

- Backbone and provider networks are called:
 - Internet Service Providers (ISPs)
- Backbone networks are often referred to as:
 - International ISPs
- Provider networks are often referred to as:
 - National ISPs
 - Regional ISPs

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Accessing the Internet - 1

- The Internet allows any user to become part of it.
- A user must be physically connected to:
 - An Internet Service Provider (ISP)
- Physical connection is normally established using:
 - Point-to-Point WAN
- Internet access methods include:
 - Using Telephone Networks
 - Using Cable Networks
 - Using Wireless Networks
 - Direct Connection to the Internet

Using Telephone Networks - 1

- Most residences and small businesses have:
 - Telephone service
- Telephone networks are already connected to:
 - The Internet
- Internet connection is created by:
 - Converting the voice line into a Point-to-Point WAN
- Two methods are used:
 - Dial-up Service
 - DSL Service

Dial-up Service

- Dial-up service uses:
 - A modem
- The modem:
 - Converts data into voice signals
- The computer software:
 - Dials the ISP
 - Creates a telephone-like connection
- Disadvantages of Dial-up:
 - Very slow speed
 - Telephone line cannot be used for voice calls simultaneously
- Suitable for:
 - Small residences
 - Occasional Internet usage

- DSL stands for:
 - Digital Subscriber Line
- Telephone companies upgraded their lines to provide:
 - Higher-speed Internet access
- DSL allows:
 - Simultaneous voice communication
 - Simultaneous data communication
- DSL is suitable for:
 - Residences
 - Small businesses

Using Cable Networks

- Many residences use:
 - Cable TV services
- Cable companies upgraded their networks and connected them to:
 - The Internet
- Internet access through cable networks provides:
 - Higher-speed connection
- Speed depends on:
 - Number of neighboring users sharing the cable

Using Wireless Networks

- Wireless connectivity has become:
 - Increasingly popular
- Homes and businesses may use:
 - Wired connections
 - Wireless connections
- Wireless WAN access allows:
 - Internet connection through wireless networks

Direct Connection to the Internet

- Large organizations can:
 - Become local ISPs
- This is achieved by:
 - Leasing a high-speed WAN
 - Connecting to a regional ISP
- Example:
 - A university with multiple campuses
 - Creates an internetwork
 - Connects the internetwork to the Internet

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Hardware and Software

- The Internet is made up of:
 - Small networks
 - Large networks
 - Connecting devices
- Simply connecting these components is not sufficient for communication.
- For communication to occur, we need:
 - Hardware
 - Software
- This is similar to a complex computation where we need:
 - A computer
 - A program